



Adapting SCADA for IP

Definition:

Adapting SCADA for IP means modifying traditional SCADA systems to use **Internet Protocol (TCP/IP)** for communication instead of serial communication.

Need for IP-based SCADA:

- Traditional SCADA uses **RS-232 / RS-485 (serial communication)**
- Limited range and speed
- No remote internet access

👉 IP-based SCADA solves these problems

Working of SCADA over IP:

- Sensors → send data to **RTU/PLC**
 - RTU/PLC → sends data via **TCP/IP network (Ethernet/Internet)**
 - Data reaches **SCADA server (Master station)**
 - User monitors through **HMI or remote device**
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Technologies Used:

- TCP/IP protocol
 - Ethernet / Wi-Fi
 - Cloud integration
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Advantages:

- 🌐 Remote monitoring and control
- ⚡ Faster data transmission
- 🔗 Easy integration with IoT systems

- 💰 Reduced wiring cost
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Disadvantages:

- 🔒 Security risks (cyber attacks)
 - 🌐 Depends on network reliability
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✅ Conclusion:

IP-based SCADA improves **flexibility, scalability, and real-time communication**, making it suitable for modern industrial and IoT applications.

👉 Tip for exam:

- Write **definition + working + advantages + conclusion**
- Draw a **simple block diagram** → extra marks 100